LOGISTIC REGRESSION PROJECT REPORT

Project: Diabetes Prediction Using Health Indicators
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1. Objective

To build and evaluate a Logistic Regression model that predicts a binary diabetes-related outcome using health indicators. Class 0 represents no diabetes; class 1 represents prediabetes or diabetes.

2. Dataset and Workflow

Dataset: BRFSS 2015 Diabetes Health Indicators. The original *Diabetes_012* target contains 0 (no diabetes), 1 (prediabetes), and 2 (diabetes). For binary classification, classes 1 and 2 were combined into the positive class.

Dataset size before duplicate removal: 253,680 rows and 22 columns. The notebook used 183,824 training rows and 45,957 testing rows, with 21 input features. No duplicate rows were found.

Workflow: load and inspect data → remove duplicates → create binary target → split data → scale features using StandardScaler → train balanced Logistic Regression (max_iter=1000) → evaluate and interpret coefficients.

3. Evaluation Results

Metric	Notebook result	Meaning
Accuracy	0.7195	Overall proportion correctly classified
Precision	0.3532	Positive predictions that were actually positive
Recall	0.7489	Actual positive cases detected
F1-score	0.4800	Balance of precision and recall
ROC-AUC	0.8053	Ability to distinguish the two classes

4. Confusion Matrix

Rows = actual class; columns = predicted class.

	Predicted 0	Predicted 1
Actual 0	TN = 27,115	FP = 10,897
Actual 1	FN = 1,995	TP = 5,950

5. Key Findings

The model achieved a ROC-AUC of 0.8053 and detected about 74.9% of positive cases. Recall was relatively high, while precision was lower, indicating that a substantial number of people predicted positive were actually in the negative class.

Largest positive coefficients reported in the notebook:

Feature	Coefficient
General Health	0.5504
BMI	0.4660
Age	0.4493
High Blood Pressure	0.3278

6. Conclusion and Limitation

This project demonstrates Logistic Regression classification using health indicators. The model achieved a ROC-AUC of 0.8053 and relatively high recall, but lower precision shows that some negative cases were incorrectly predicted as positive.

This model is for educational use only and must not be used for medical diagnosis. The assignment handout specifies the Pima Indians Diabetes dataset, while this notebook uses BRFSS 2015; confirm that the dataset substitution is acceptable before submission.